



On the use of VAE for ecomaterial selection with applications in structural and aircraft design optimization

Prof. Joseph Morlier



PEML25- Lightning talks

Co-authors and MS students ;)
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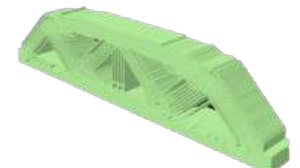
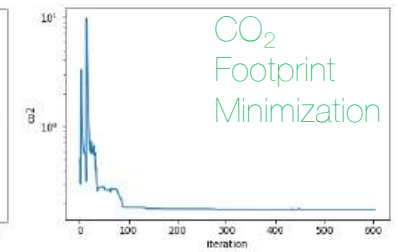
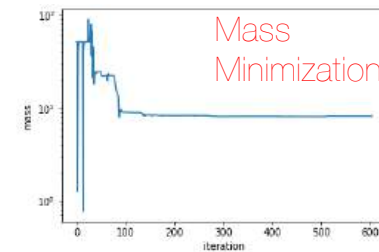
Aircraft Design

Fluid x control x physics x applied maths x structures & materials

In AD: Strong coupling
Between
Disciplines (MDO)

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
Avionics group	1	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•
Electrical group	2	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•
Escape system	3	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•
Armament	4	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•
Landing gear	5	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•
Hydraulics group	6	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•
Flight control system	7	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•
Environment and control system	8	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•
Power plant group	9	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•
Fatigue group	10	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•
Aero elastic group	11	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•
Stress group	12	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•
Materials group	13	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•
Fly wheel	14	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•
Empennage group	15	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•
Rear fuselage	16	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•
Fuselage group	17	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•

Topology x Material x Process



$$\text{Range} = V t_f = V \times \underbrace{\left(\frac{L}{D} \right)}_{\text{aircraft designer}} \times \underbrace{I_{sp}}_{\text{propulsion system designer}} \times \underbrace{\ln \left(\frac{W_i}{W_f} \right)}_{\text{structural designer}}$$

The main problem:
introducing *discrete* material database in a
continuous gradient based optimization

HOW?

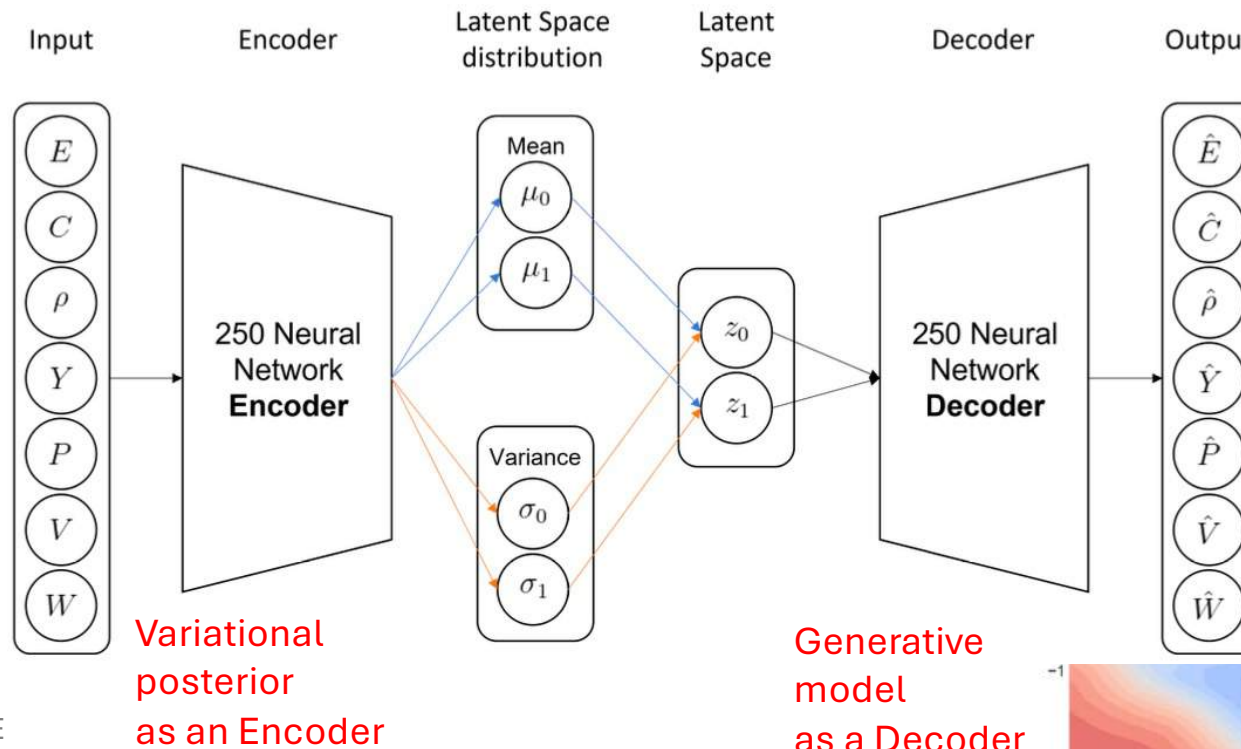
Variational AutoEncoder (VAE):
class of probabilistic models that find latent, low-
dimensional representations of data.

Kingma, D. P., & Welling, M. "Auto-Encoding Variational Bayes", 2013.

From catalog to *differentiable* material representation

Table 1 A curated subset of material training

Material	Class	E [Pa]	C
A286 iron	Steel	$2.01\text{E}+11$	5.
AISI 304	Steel	$1.90\text{E}+11$	2.
Gray cast iron	Steel	$6.62\text{E}+10$	6.
3003-H16	Al alloy	$6.90\text{E}+10$	2.
5052-O	Al alloy	$7.00\text{E}+10$	2.
7050-T7651	Al alloy	$7.20\text{E}+10$	2.
Acrylic	Plastic	$3.00\text{E}+09$	2.
ABS	Plastic	$2.00\text{E}+09$	2.
PE HD	Plastic	$1.07\text{E}+09$	2.



Framework: PyTorch VAE

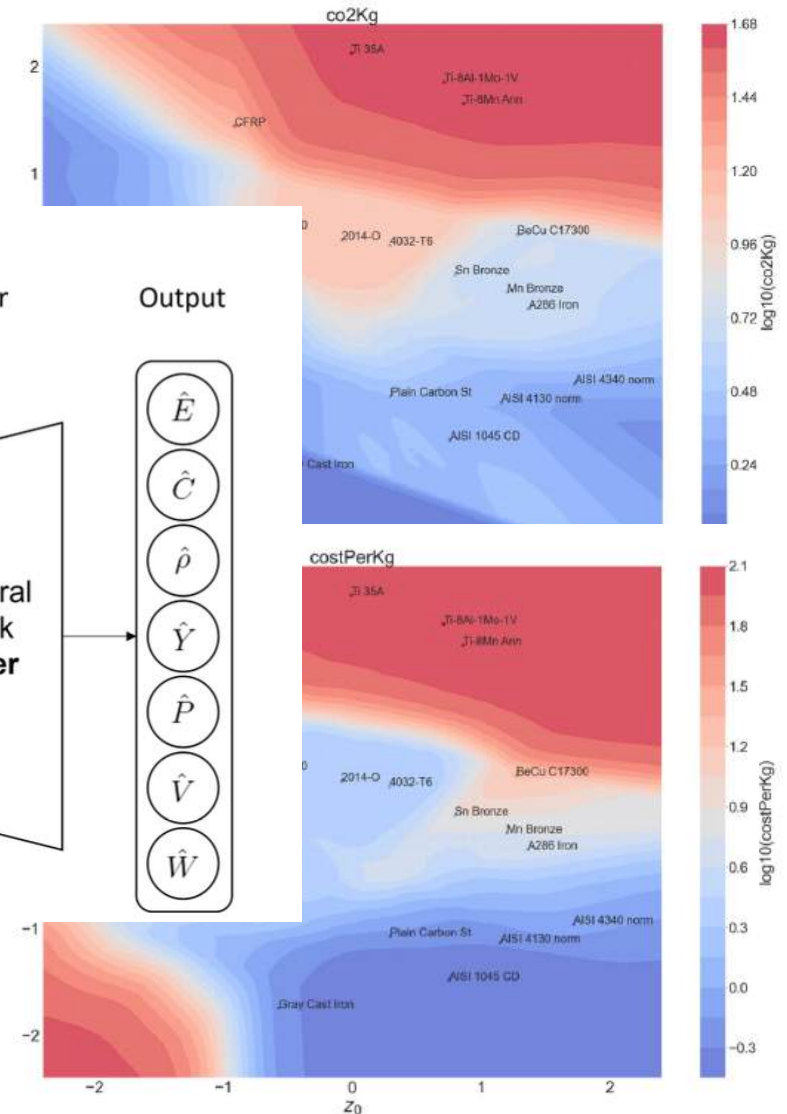
Network architecture: Fully connected with 250 neurons in encoder and decoder

Latent space: 2D (z_0, z_1)

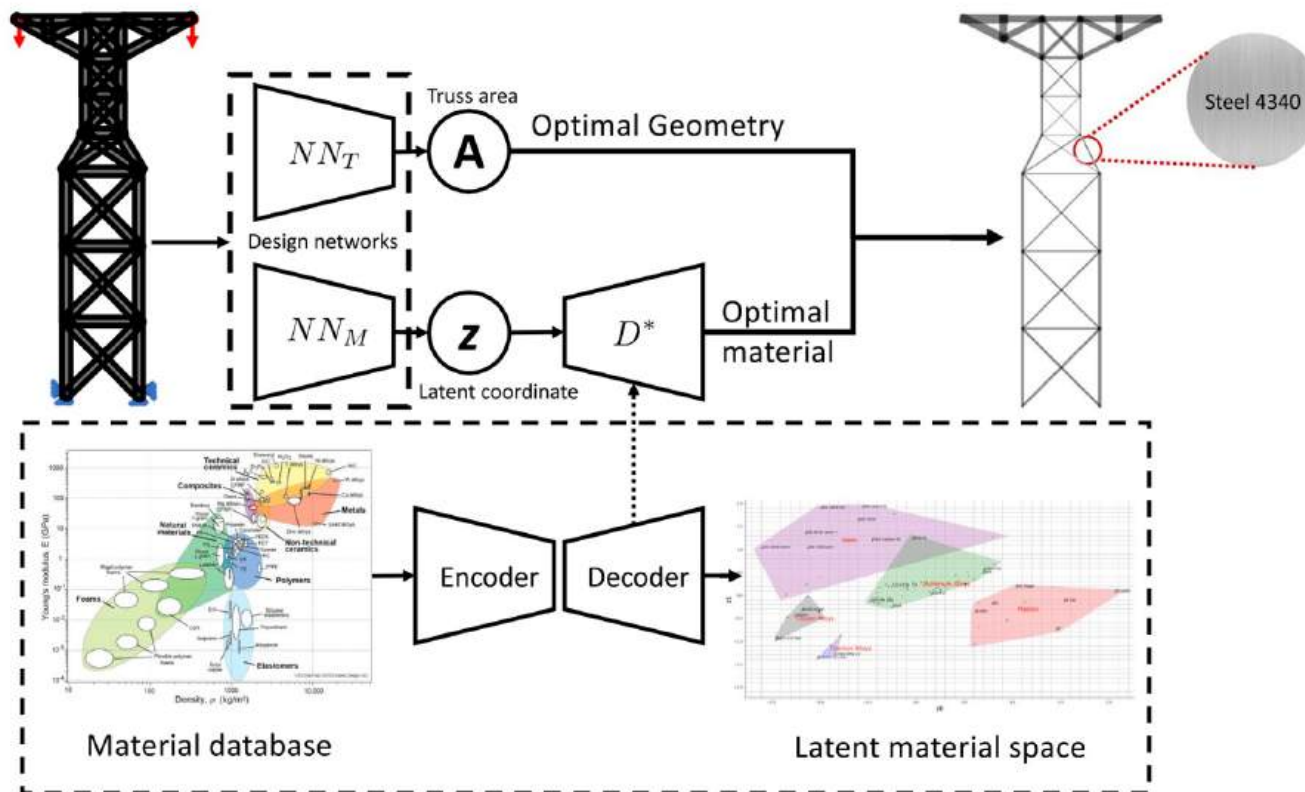
Inputs normalized to $[0, 1]$; outputs rescaled post-decoding

Training time: 3 minutes on a CPU (50,000 iterations)

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Everything is continuous here !



Chandrasekhar, A., Sridhara, S., & Suresh, K. (2022). Integrating material selection with design optimization via neural networks. *Engineering with Computers*, 38(5), 4715-4730.

LCA & eco selection

- Material
- Process
- from cradle to grave
- ...

Water withdrawal



Generation of waste



Use recycled:
Fibers
Resin
Metals
Reuse & Repair

Carbon Footprint



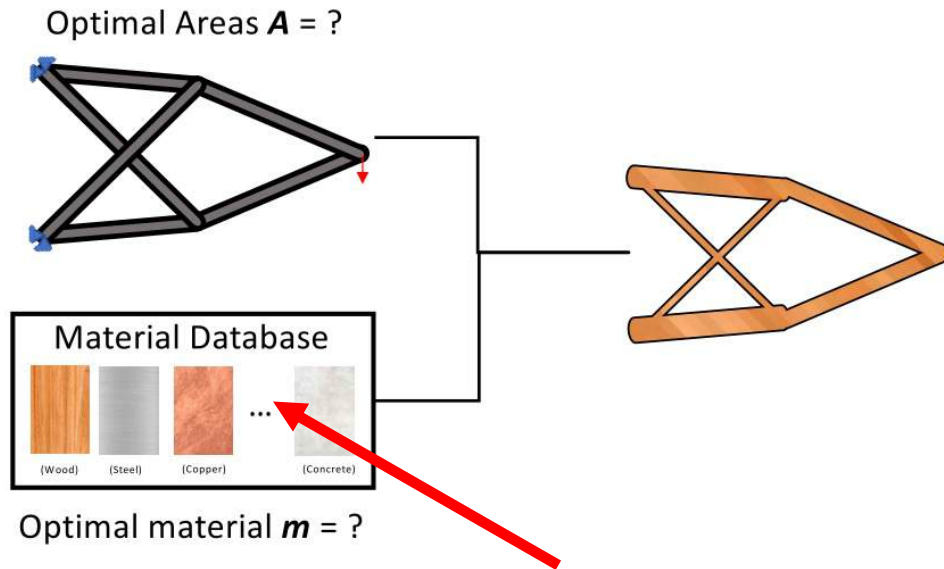
Energy requirement



Yepes Llorente, L., Morlier, J., Sridhara, S., & Suresh, K. (2024). A hybrid machine learning and evolutionary approach to material selection and design optimization for eco-friendly structures. *Structural and Multidisciplinary Optimization*, 67(5), 69.

Testcase #1 : structural optimization

topology is fixed !

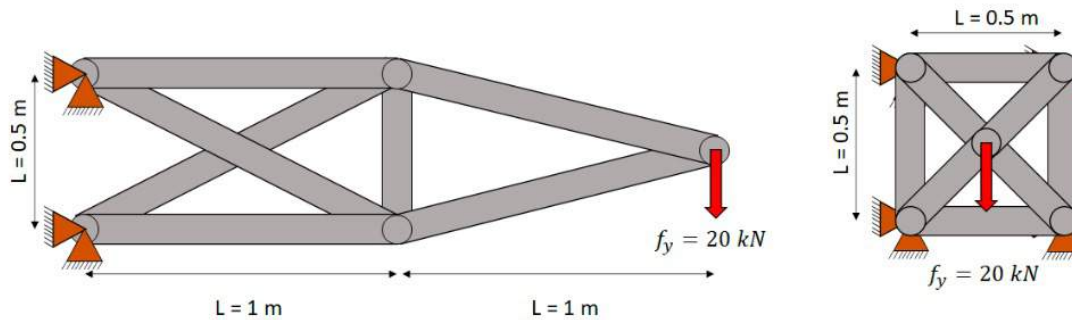


First: Reduced database of
material's candidates
Using our engineer's experience

Objective	Equation
Compliance	$\mathbf{f}^\top \mathbf{u}(\mathbf{A}, \hat{E})$
Mass	$\hat{\rho} \sum_{k=1}^N A_k L_k$
Cost	$\hat{\rho} \hat{C} \sum_{k=1}^N A_k L_k$
CO2	$\hat{\rho} \hat{P} \sum_{k=1}^N A_k L_k$
Energy	$\hat{\rho} \hat{V} \sum_{k=1}^N A_k L_k$
Water	$\hat{\rho} \hat{W} \sum_{k=1}^N A_k L_k$

Table B2: Available objective functions

Environmentally friendly 3D truss

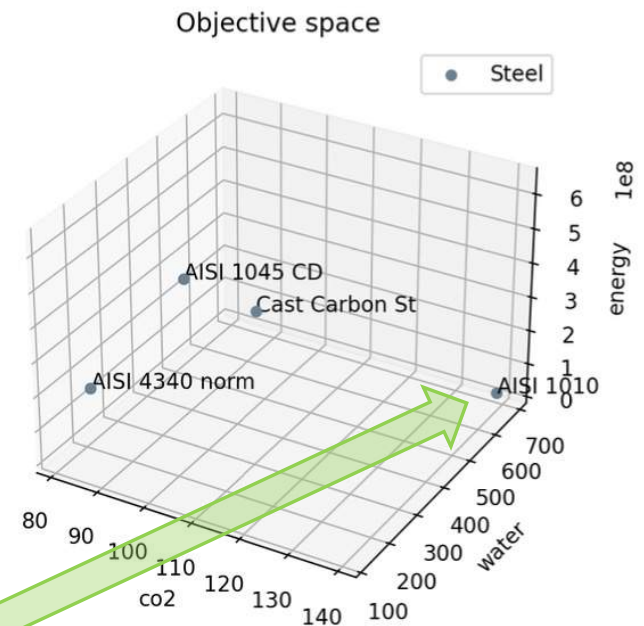


Being P_{min} , W_{min} and V_{min} the minimum value of CO_2 , water and energy achievable respectively, the index is defined as:

$$H = \frac{1}{3} \frac{P - P_{min}}{P_{min}} + \frac{1}{3} \frac{W - W_{min}}{W_{min}} + \frac{1}{3} \frac{V - V_{min}}{V_{min}} \quad (7)$$

In such a way that the perfect material with minimum consumption would yield $H = 0$. To take into account the cost of the structure as well, both values were calculated for each solution and presented in Table 10.

Material	Cost	Env. Consumption
AISI 1045 CD	18.63 \$	9.60 [-]
Cast Carbon St	28.68 \$	7.85 [-]
AISI 4340 norm	31.91 \$	2.28 [-]
AISI 1010	24.04 \$	1.73 [-]



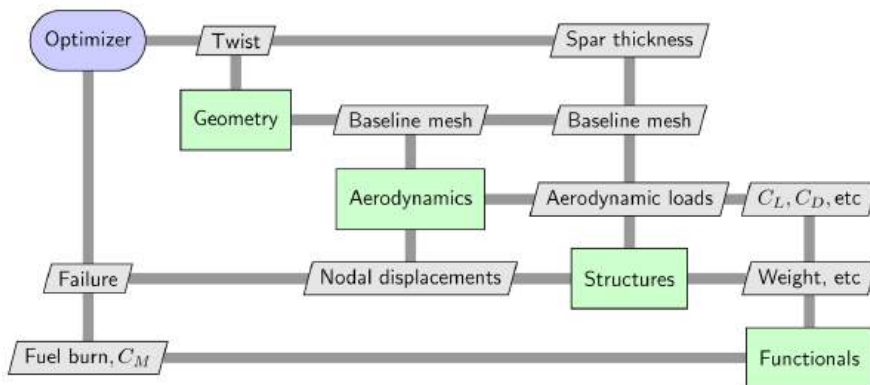
Testcase #2: Aircraft wing optimization

Topology is free in the MDO problem

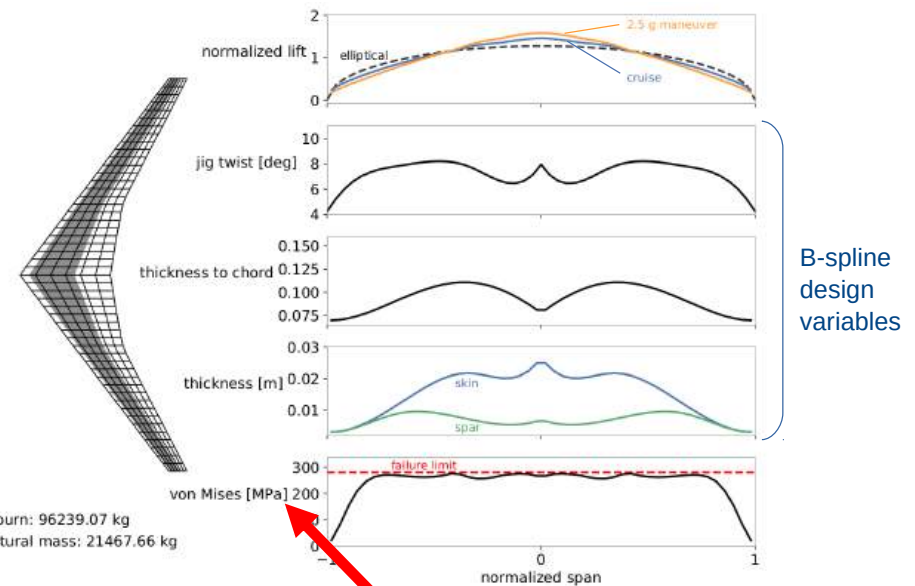
Results

This is MDO

Discipline Structure is coupled to
Discipline Aerodynamic !



Minimize fuel burn



fuel burn: 96239.07 kg
structural mass: 21467.66 kg

Wrt

L=W

Admissible stress

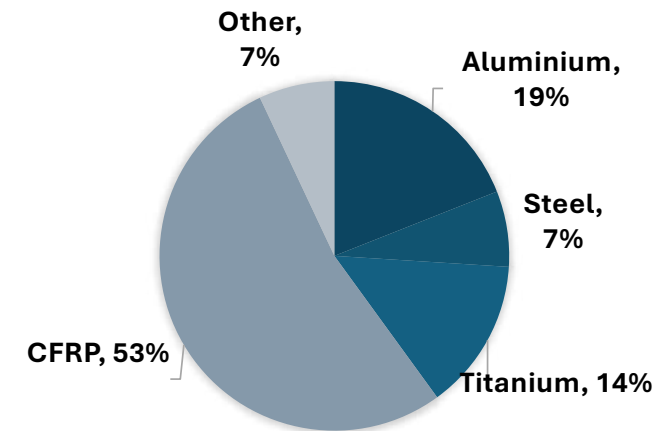
Etc...

Jasa, J. P., Hwang, J. T., & Martins, J. R. (2018). Open-source coupled aerostructural optimization using Python. *Structural and Multidisciplinary Optimization*, 57(4), 1815-1827.
Chauhan, S. S., & Martins, J. R. (2018, September). Low-fidelity aerostructural optimization of aircraft wings with a simplified wingbox model using OpenAeroStruct. In *International conference on engineering optimization* (pp. 418-431). Cham: Springer International. PEML25

Testcase #2: Aircraft wing optimization with *a single material*



Carbon fiber airframe: e.g. a350



Real aircrafts are multimaterials-based:

How to find new hybrid eco-materials?

Solve a MultiObjective Optimization (MOO)

Minimize:

cost (C), density-to-stiffness ratio (r), energy (V) and waste (W)

Maximize: yield strength (Y)

What VAE can do ?

- Differentiable optimization: Gradients of decoded properties w.r.t. latent variables

$$\frac{\partial \hat{E}}{\partial z_0}, \quad \frac{\partial \hat{C}}{\partial z_1}, \quad \text{etc.}$$

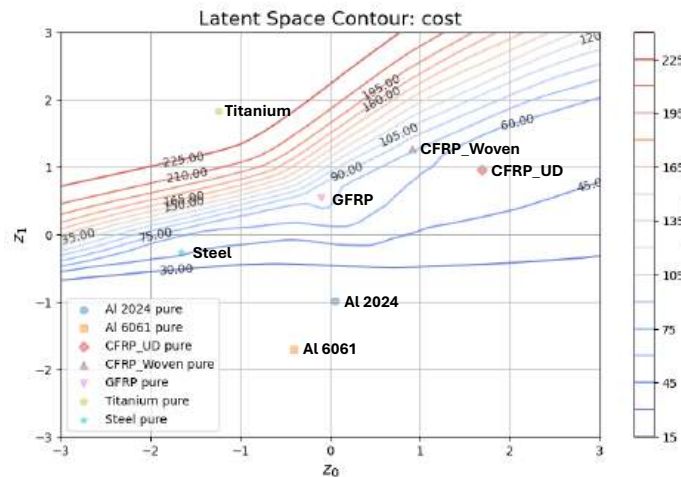
multimaterials

- Interpolation: New combinations of materials with hybrid properties
- Dimensionality reduction: Simplifies optimization over material properties

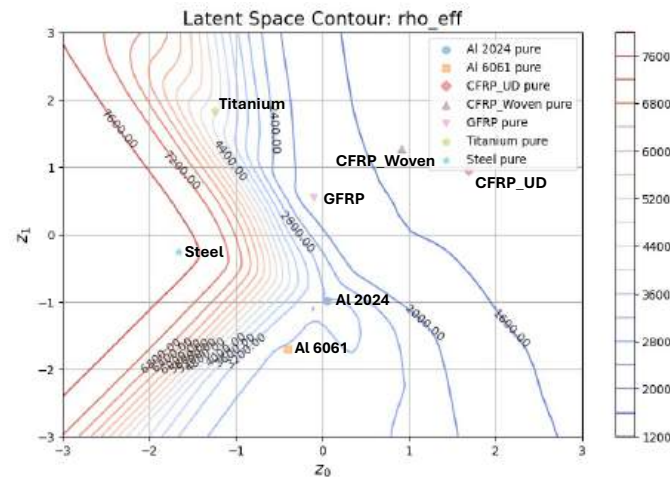
Property	Al 2024	Al 6061	CFRP-UD	CFRP-Woven	GFRP	Titanium	Steel
ρ [kg/m ³]	2765	2710	1565	1575	1850	4430	7740
Cost [€/kg]	17.28	16.20	55.28	78.00	76.80	236.00	45.57
Energy [MJ/kg]	872.5	890.0	1091.3	1091.3	311.7	6875.0	411.3
Waste [kg/kg]	0.20	0.20	0.50	0.50	2.00	1.35	0.25
E [GPa]	73.85	68.30	141.50	65.70	33.30	114.50	200.00
σ_y [MPa]	331.0	127.5	1955.0	768.5	368.5	848.0	698.5

Feed Multiple-Gradient Descent Algorithm (MGDA)

J.-A. Désidéri, A. Minelli, and A. Zerbinati, "A cooperative algorithm for multi- objective optimization: multiple-gradient descent algorithm (mgda)," in 4th Inverse Problems, Design and Optimization Symposium (IPDO-2013), Albi, France, Jun. 2013.

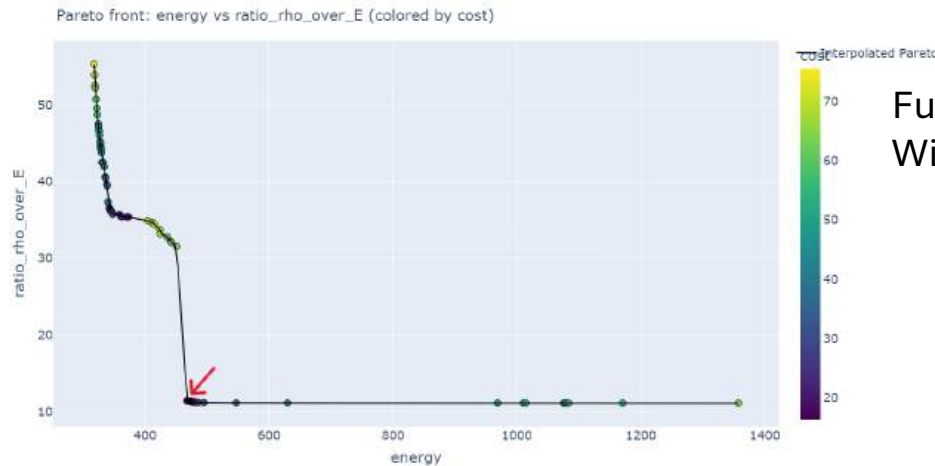


- Al 2024 and Al 6061 lie below **30 €/kg** contours
- Reflects true costs: **17.28 / 16.20 €/kg**
- Latent space preserves **relative material cost**



- Both CFRP's sit just above **1600 kg/m³** contour
- Matches true densities: **1565 / 1575 kg/m³**
- Latent space reflects **physical accuracy**

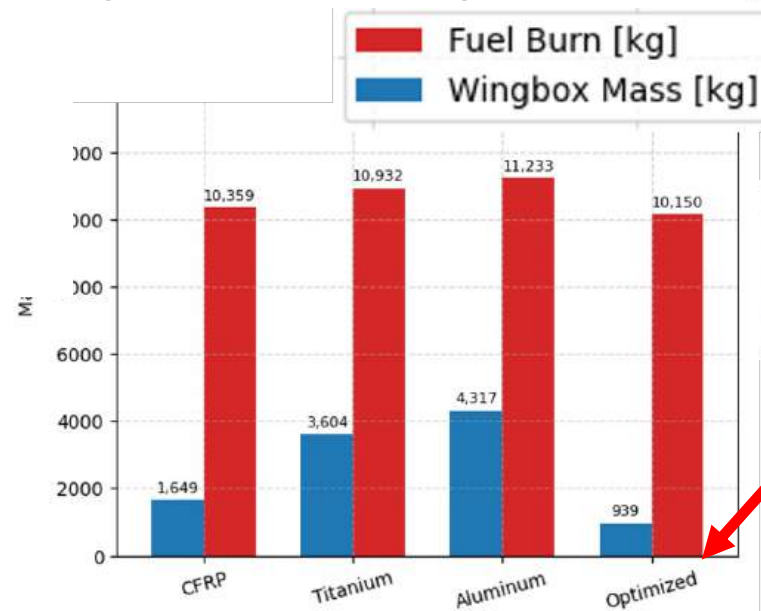
Testcase #2: MGDA results



A320 Improvements - Optimized vs:

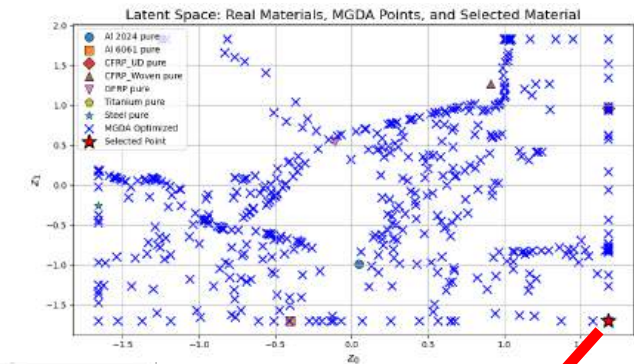
- CFRP:
 - Fuel ↓ 2.02%
 - Wingbox Mass ↓ 43.05%
- Titanium:
 - Fuel ↓ 7.16%
 - Wingbox Mass ↓ 73.94%
- Aluminum:
 - Fuel ↓ 9.65%,
 - Wingbox Mass ↓ 78.24%

Fuel consumption: 10,149.76 kg
Wingbox mass: 939.34 kg



Property	Value	Unit
Energy consumption per unit mass	467.31	MJ/kg
Density-to-stiffness ratio (ρ/E)	11.40	kg/(m ³ ·GPa)
Effective density	1790.97	kg/m ³
Cost per unit mass	16.24	€/kg
Waste generation	0.20	kg/kg
Yield stress	1129.31	MPa
Young's modulus	157.12	GPa

Fictive new High-modulus low Environmental impact hybrid composite is probably the nearest real-world approximation.



Another solution... Another story

- *material selection as categorical variables ?*



What kind of
variable types
are available in
SMT 2.0 ?

Continuous

Discrete

Categorical

Hierarchical

Mixed

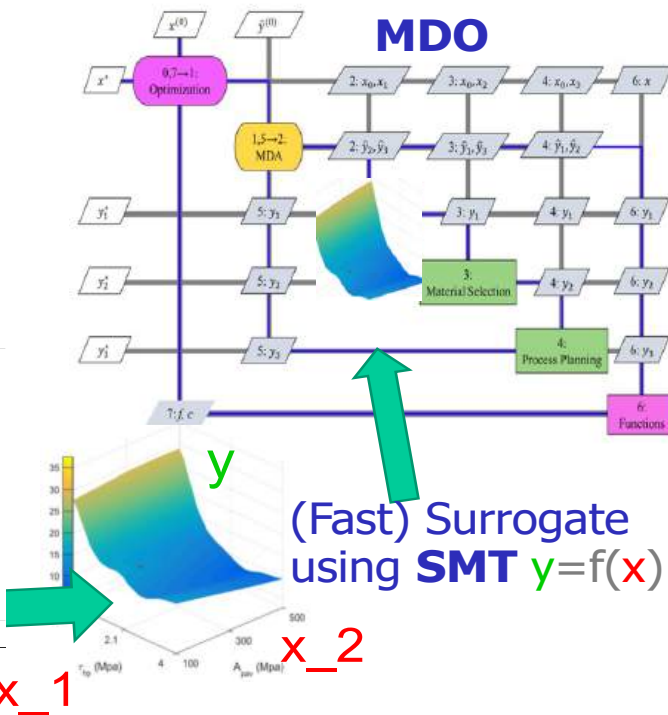
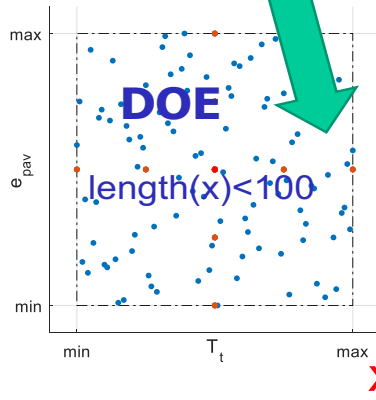
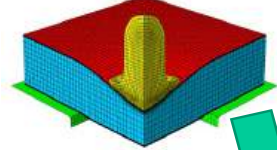
Bouhlel, M. A., Hwang, J. T., Bartoli, N., Lafage, R., Morlier, J., & Martins, J. R. (2019). A Python surrogate modeling framework with derivatives. *Advances in Engineering Software*, 135, 102662.

Saves, P., Lafage, R., Bartoli, N., Diouane, Y., Bussemaker, J., Lefebvre, T., ... & Martins, J. R. (2024). SMT 2.0: A Surrogate Modeling Toolbox with a focus on hierarchical and mixed variables Gaussian processes. *Advances in Engineering Software*, 188, 103571.

Interested in Surrogate modeling? GP/BO?

Big picture?

(Expensive)
simulation



(Fast) Surrogate
using **SMT** $y=f(x)$



<https://github.com/SMTorg/smt>